Concrete Antifer Cubes

Executive Summary

Antifer cubes are massive grooved unreinforced concrete armour units used in rubble-mound breakwaters and related maritime protection works. They were developed for the Port of Antifer, Normandy, in the early 1970s, as a robust improvement over the plain concrete cube. The geometry remains compact and structurally conservative, but adds four lateral grooves and controlled tapering to improve hydraulic behaviour, roughness, interlock and resistance to face-to-face rearrangement.

This repository contains a Python production tool that generates a closed vectorial Antifer armour-unit solid from an input unit weight W and concrete unit weight W_c. The tool exports the same validated geometry to IFC, STL, OBJ and DXF. It is a geometric CAD/BIM production and documentation tool; hydraulic stability design remains a separate project-specific task.

Two dimensional conventions are implemented.

Convention	Volume relation
Pita (1986)	$V/H^3 = 1.024700000000$
Carvalho (2026)	$V/H^3 = 1.000000000000$

Pita (1986). Purpose: historical proportional Antifer reference. Reference-height meaning: drawing height used in the historical coefficient set.

Carvalho (2026). Purpose: volume-normalised CAD/BIM reference. Reference-height meaning: $H = D_n = V^{1/3}$.

The same vectorial topology is used in both conventions: a tapered body, no top bevel, four lateral plan chamfers and four single circular-arc side grooves. The Carvalho convention is obtained by horizontally normalising the Pita plan and groove coefficients by a common factor so that the final non-dimensional volume coefficient becomes unity.

Repository Scope

This repository is concerned with the reproducible geometric reconstruction of Antifer cubes. It does not replace hydraulic model testing, breakwater stability design, overtopping analysis, toe design, crest design or construction-method verification.

The main outputs are:

Output	Description
IFC	IFC4 `IfcFacetedBrep` model with property and quantity data
STL	closed triangular surface mesh in metres
OBJ	indexed triangular surface mesh in metres
DXF	AutoCAD 2018 / AC1032 file with an ACIS-backed `3DSOLID`
TXT	`output.txt` calculation and validation report

Historical and Engineering Context

The Antifer cube is an improved massive concrete cube. It should not be treated as a slender high-interlock unit. Its engineering value lies in combining hydraulic improvement with structural robustness.

The unit became especially relevant after the structural-integrity problems observed in some slender concrete armour units during the late 1970s and early 1980s. Antifer cubes retain large resistant concrete sections and avoid slender arms, reducing the risk of bending-sensitive prototype breakage. This robustness comes at the cost of relatively high concrete consumption compared with modern monolayer armour units.

Representative contexts in which Antifer units have been used or studied include:

Case	Main relevance
Port of Antifer	origin and prototype context
Sines West Breakwater	post-Dolos failure rehabilitation
Zeebrugge	prototype overtopping and run-up
Douro Bar / Cabedelo do Douro	tests, high-density units, monitoring
Leixoes Outer Breakwater	contemporary Portuguese port extension
Lajes das Flores	Atlantic island emergency/reconstruction
Mohammedia	international severe-exposure context
Brunei	manufacture, curing and placement

At Sines, this concerns repair and rehabilitation after Dolos armour-layer failure. Zeebrugge provides prototype measurements; Douro includes model testing. Lajes das Flores concerns island-port emergency protection and reconstruction. The Brunei case documents manufacture, curing and placement practice.

These examples are engineering context only. They do not define a universal stability coefficient.

Geometry and Dimensional Convention

Adopted Production Geometry

Item	Adopted convention
Vertical reference	H is the total unit height
Bottom section	A is the bottom bounding width at $z = 0$
Top section	B is the top bounding width at $z = H$
Top face	single horizontal plane; no top bevel
Chamfers	D is a lateral plan-chamfer offset
Grooves	four identical side grooves
Groove type	one continuous circular arc per side
Groove radius	R, with $r = R/H$
Groove centre offset	E, with $e = E/H$, outside the side face
Generated groove depth	$C_{\text{arc}} = R - E$; $c_{\text{arc}} = r - e$
Groove half-opening	$Q = \sqrt{R^2 - E^2}$; $q = Q/H = \sqrt{r^2 - e^2}$

Geometry source: pure vectorial B-rep generation; no imported template geometry.

Master Coefficient Table

All Pita and Carvalho coefficients used by the production script are consolidated here. Later sections refer back to this table instead of repeating the same coefficient set.

Quantity	Pita (1986)	Carvalho (2026)
K_V	1.024700000000	1.000000000000
f	1.000000000000	0.987874174182
$a = A/H$	1.086000000000	1.072831353161
$b = B/H$	1.005000000000	0.992813545052
$d = D/H$	0.024000000000	0.023708980180
$r = R/H$	0.121500000000	0.120026712163
$e = E/H$	0.025906459610	0.025592322393
$c_{\text{arc}} = r - e$	0.095593540390	0.094434389770
$q = \sqrt{r^2 - e^2}$	0.118705961731	0.117266553915
$2q$	0.237411923462	0.234533107831

Coefficient definitions:

Quantity	Meaning
K_V	final volume coefficient, V/H^3
f	horizontal factor applied to Pita coefficients
$a = A/H$	bottom width coefficient
$b = B/H$	top width coefficient
$d = D/H$	lateral chamfer coefficient
$r = R/H$	groove-radius coefficient
$e = E/H$	outside groove-centre offset
$c_{\text{arc}} = r - e$	generated circular-arc groove-depth coefficient
$q = \sqrt{r^2 - e^2}$	groove half-opening coefficient
$2q$	full groove-opening coefficient

The historical Pita notation often reports $C/H = 0.095600$. The production geometry uses the volume-consistent circular-arc value $c_{\text{arc}} = 0.095593540390$, which is the generated depth obtained from $C_{\text{arc}} = R - E$.

Algebraic Volume Definition

The generated unit volume is computed from a simple geometric balance.

$$\frac{V}{H^3} = K_{\text{frustum}} - K_{\text{chamfers}} - K_{\text{grooves}}$$

The components are:

Component	Non-dimensional expression
Gross tapered square frustum	$K_{\text{frustum}} = (a^2 + ab + b^2)/3$

Component	Non-dimensional expression
Four corner chamfers	$K_{\text{chamfers}} = 2d^2$
One circular groove	$A_g = r^2 \arccos(e/r) - e\sqrt{r^2 - e^2}$
Four side grooves	$K_{\text{grooves}} = 4A_g$
Final coefficient	$K_V = K_{\text{frustum}} - K_{\text{chamfers}} - K_{\text{grooves}}$

Component meanings:

- **Gross tapered square frustum:** volume before groove and chamfer removals.
- **Four corner chamfers:** four triangular plan removals carried through the height.
- **One circular groove:** circular-segment area removed by one side groove.
- **Four side grooves:** total groove subtraction.
- **Final coefficient:** final value of V/H^3.

Demonstration of the Square-Frustum Formula

Let a square frustum have height H, lower width A and upper width B. At a vertical coordinate z, measured from the bottom, the linearly interpolated side length is:

$$s(z) = A + \frac{B - A}{H} z.$$

The horizontal area at level z is:

$$S(z) = s(z)^2.$$

The frustum volume is therefore:

$$V_{\text{frustum}} = \int_0^H \left(A + \frac{B - A}{H} z \right)^2 dz.$$

Expanding and integrating:

$$V_{\text{frustum}} = \int_0^H \left[A^2 + 2A \frac{B - A}{H} z + \left(\frac{B - A}{H} \right)^2 z^2 \right] dz,$$

$$V_{\text{frustum}} = A^2 H + A(B - A)H + \frac{(B - A)^2}{3} H.$$

Collecting terms gives:

$$V_{\text{frustum}} = \frac{H}{3} \left(A^2 + AB + B^2 \right).$$

Using $a = A/H$ and $b = B/H$:

$$\frac{V_{\text{frustum}}}{H^3} = \frac{a^2 + ab + b^2}{3}.$$

This is the gross body used before subtracting the four lateral corner chamfers and the four circular side grooves.

Compact Numerical Balance

Component	Pita (1986)	Carvalho (2026)
Gross frustum	1.093617000000	1.067255782180
Chamfers	0.001152000000	0.001124231482
Grooves	0.067765000000	0.066131550698
Final V/H^3	1.024700000000	1.000000000000

For Pita, the one-groove area is:

$$A_g = \frac{1.092465000000 - 1.024700000000}{4} = 0.016941250000$$

With r = 0.121500000000, this value is obtained by solving:

$$A_g = r^2 \arccos\left(\frac{e}{r}\right) - e\sqrt{r^2 - e^2}$$

which gives:

$$e = 0.025906459610, \qquad q = 0.118705961731, \qquad c_{\text{arc}} = 0.095593540390$$

For Carvalho, all Pita horizontal plan and groove coefficients are multiplied by:

$$f = \sqrt{\frac{1.000000000000}{1.024700000000}} = 0.987874174182$$

Because the vertical reference height remains H, the non-dimensional area terms scale with f^2, giving $K_V = 1.000000000000$.

Dimensional Scaling from Input Weight

The script uses the input unit weight W and concrete unit weight W_c to compute the unit volume:

$$V = \frac{W}{W_c}$$

The selected dimensional convention then gives the reference height:

Convention	Height relation
Pita (1986)	$H = (V/1.024700000000)^{1/3}$
Carvalho (2026)	$H = V^{1/3} = D_n$

Once H is known, all absolute dimensions are obtained by multiplying the coefficient table values by H. For example, $A = aH$, $B = bH$, $D = dH$, $R = rH$ and $E = eH$.

Nominal Diameter

The Van der Meer nominal diameter is:

$$D_n = V^{1/3}$$

The relation between D_n and the drawing height depends on the selected convention:

Convention	Relation
Pita (1986)	$D_n = (1.024700000000)^{1/3} H = 1.008166460709H$
Carvalho (2026)	$D_n = H$

This distinction is important when hydraulic sizing is expressed through D_n but CAD geometry is generated through a drawing height H.

Horizontal Section and Groove Construction

A horizontal section has local half-width $h = w/(2H)$, where w is the non-dimensional section width. The east-side groove can be written in local non-dimensional coordinates as:

$$x(y) = h + e - \sqrt{r^2 - y^2}, \qquad -q \le y \le q$$

Key points are:

Feature	Coordinate or relation
Groove mouth	$y = \pm q$, $x = h$
Deepest groove point	$y = 0$, $x = h + e - r = h - c_{\text{arc}}$
Other sides	obtained by 90-degree rotations
Corner chamfers	four lateral plan chamfers using offset d

This section definition is used consistently in IFC, STL, OBJ and DXF exports.

Hydraulic Stability Interpretation

Antifer hydraulic stability is not governed by geometry alone. It depends strongly on the armour-layer system: placement method, packing density, slope, wave climate, storm duration, underlayer, toe support, crest details and damage criterion.

The Hudson formula remains a common preliminary sizing relation:

$$W = \frac{\gamma_c H_D^3}{K_D \Delta^3 \cot \alpha}, \qquad \Delta = \frac{\gamma_c}{\gamma_w} - 1$$

The stability number formulation is often more convenient for comparison:

$$N_s = \frac{H_s}{\Delta D_n}$$

For regular two-layer cubes and Antifer-calibrated formulae, a Van der Meer-type empirical structure may be used:

$$N_s = \left(k_1 \frac{N_{od}^{k_2}}{N_z^{k_3}} + k_4 \right) s_{0m}^{-k_5}$$

Typical coefficient sets are:

Armour unit and slope	k_1	k_2	k_3	k_4	k_5
Regular cube, $\cot\alpha = 1.5$	6.700	0.400	0.300	1.000	0.100
Regular cube, $\cot\alpha = 2.0$	7.374	0.400	0.300	1.101	0.100
Antifer, $\cot\alpha = 1.5$	6.951	0.443	0.291	1.082	0.082
Antifer, $\cot\alpha = 2.0$	6.138	0.443	0.276	1.164	0.070

These formulae are preliminary design tools, not substitutes for project-specific verification. Reported K_D values for Antifer vary widely because placement method and damage definition dominate the result.

Placement, Packing and Failure Modes

The same Antifer geometry can perform differently depending on armour-layer architecture.

Placement tendency	Main benefit
regular placement	higher displacement stability
irregular placement	greater roughness and reduced overtopping
dense packing	higher contact restraint
open packing	higher porosity and roughness

Placement tendency	Main risk
regular placement	greater reflection and overtopping in some cases
irregular placement	lower displacement stability
dense packing	reduced porosity and possible paving behaviour
open packing	greater movement risk

Relevant failure and degradation mechanisms include:

Mechanism	Description
rocking	repeated angular motion; no extraction
displacement	block leaves its stable position
chain failure	block movement destabilises adjacent units
smoothing or paving	more regular, less rough armour surface
concrete cracking	thermal, shrinkage, handling, impact or fatigue
toe or underlayer failure	support loss; progressive armour instability
overtopping rear damage	transmitted water damages crest or rear side

A defensible Antifer design must therefore consider the full armour system, not only the isolated concrete unit.

Manufacture, Materials and Quality Control

Antifer units are normally manufactured as plain unreinforced concrete blocks. The main production advantages are simple formwork, robust geometry and straightforward inspection. The main risks are thermal cracking, honeycombing, density variation and handling damage.

Aspect	Typical practice or control
Concrete	normally unreinforced; low hydration heat
Strength class	commonly C25/30 in manual-type references
Formwork	four-face mould; typically open top and bottom
Casting	top casting with vibration
Curing	prolonged moist curing, especially for large units
Handling	controlled lifting, rotation and storage procedures
Mass control	verify W = Wc V
Dimensional control	verify mould dimensions and generated volume
Rejection logic	reject the defects listed below

Low heat of hydration is recommended. Reject units with significant cracks, honeycombing, cold joints or excessive corner loss.

A dimensional error is a stability error. A density error is also a stability error. For CAD/BIM production, a volume error in the digital solid becomes a procurement, fabrication and quality-control error.

Python Script Installation and Usage

Requirements

Python 3.10 or newer is recommended. Tkinter must be available; on standard Windows Python installations it is normally included.

Install the runtime packages with:

```
python -m pip install ezdxf trimesh pillow
```

`pillow` is recommended for broad Windows and PyInstaller compatibility with optional `ezdxf` drawing add-ons.

Windows Virtual Environment

Create a virtual environment:

```
py -m venv .venv
```

Activate it:

```
.\.venv\Scripts\activate
```

Upgrade `pip`:

```
python -m pip install --upgrade pip
```

Install the dependencies:

```
python -m pip install ezdxf trimesh pillow
```

For standalone executable compilation, also install PyInstaller:

```
python -m pip install pyinstaller
```

Running the Application

Run the graphical application with:

```
python script_final_production.py
```

Typical workflow:

1. Enter the Antifer unit weight W in kN.
2. Enter the concrete unit weight W_c in kN/m3.
3. Select Pita (1986) or Carvalho (2026).
4. Select the output folder.
5. Review the calculation and validation preview.
6. Press **Generate IFC + STL + OBJ + DXF**.

Example output names:

```
antifer_pita1986_500kN_Wc24kNm3.ifc
antifer_pita1986_500kN_Wc24kNm3.stl
antifer_pita1986_500kN_Wc24kNm3.obj
antifer_pita1986_500kN_Wc24kNm3.dxf
output.txt
```

Viewing the Exported Files

Format	Recommended inspection
IFC	IFC/BIM viewers supporting faceted B-rep geometry
STL	mesh viewers or CAD applications with mesh import
OBJ	mesh viewers or modelling tools
DXF	AutoCAD/ZWCAD-type CAD inspection as ACIS-backed `3DSOLID`
TXT	direct review of dimensions and validation values

All exported geometry is dimensional and written in metres. The exported XY origin is placed at the homogeneous centre of mass, while the exported Z datum remains the physical bottom level z = 0.

Standalone Executable Compilation

A PyInstaller spec file is recommended for repeatable Windows production builds:

```
pyi-makespec --onefile --windowed --name script `
  --collect-all ezdxf script_final_production.py
pyinstaller --clean --noconfirm script.spec
```

The compiled executable is written to the `dist` folder.

Verification Checklist

Before issuing geometry for technical use, check:

Check	Required result
selected convention	Pita or Carvalho coefficients match the master table
top face	one flat horizontal face at z = H, with no top bevel
chamfers	four lateral plan chamfers only
grooves	one circular arc per side, radius R, centre offset E
closed shell	every B-rep edge shared by exactly two faces
volume	generated volume matches the selected K_V
IFC/STL/OBJ/DXF	all formats generated from the same B-rep source
output report	dimensions and validation values in `output.txt`

Limitations

This repository does not provide:

- new hydraulic model tests;
- prototype monitoring data;
- finite-element structural-integrity analysis;

- a complete breakwater design method;
- a universal Antifer K_D;
- project-specific wave, overtopping, toe, crest or underlayer verification.

Critical works should be verified through physical modelling or equivalent project-specific evidence.

Conclusions

The production geometry is a reproducible algebraic reconstruction of an Antifer cube as a square frustum with four lateral plan chamfers and four circular side grooves. Pita preserves the historical volume coefficient $V/H^3 = 1.024700000000$. Carvalho keeps the same topology but horizontally normalises the plan and groove coefficients so that $V/H^3 = 1.000000000000$ and $H = D_n$.

The geometric definition is necessary for reliable CAD/BIM production, drawing control, volume calculation and quality assurance. It does not remove the need for hydraulic verification. Antifer stability remains governed primarily by armour-layer placement, packing density, wave climate, slope, storm duration, toe support, construction tolerance and accepted damage.

Dedication

This work is respectfully dedicated to the memory of **Carlos Alberto Roldão Maia Pita** (1950-2002), Portuguese civil engineer and specialist in maritime hydraulics and rubble-mound breakwater design. His LNEC *Memória n.º 670 — Dimensionamento Hidráulico do Manto Resistente de Quebra-Mares de Talude* remains an important technical reference for Portuguese maritime engineering and provides a key historical basis for the geometric interpretation of Antifer cubes.

References

1. Pita, C. A. R. M. *Dimensionamento Hidraulico do Manto Resistente de Quebra-Mares de Talude*. Laboratorio Nacional de Engenharia Civil, Memoria n.º 670, Lisboa, 1986.
2. BSI. *BS 6349-7:1991 Maritime structures - Part 7: Guide to the design and construction of breakwaters*, with 2010 corrigenda.
3. CIRIA; CUR; CETMEF. *The Rock Manual: The use of rock in hydraulic engineering*, 2nd ed., 2007.
4. U.S. Army Corps of Engineers. *Coastal Engineering Manual*. Engineer Manual EM 1110-2-1100, U.S. Army Corps of Engineers, Washington, D.C., 2002.
5. Yagci, O.; Kapdasli, S.; Cigizoglu, H. K. "The stability of the antifer units used on breakwaters in case of irregular placement", *Ocean Engineering*, 31, 1111-1127, 2004.
6. Frens, A. B. *The Impact of Placement Method on Antifer-block Stability*. MSc thesis, Delft University of Technology, 2007.
7. Neves, M. G.; Silva, L. G.; Reither, S. *Stability and overtopping of rubble-mound breakwaters: influence of the placement method and density of cubic blocks*. LNEC / PIANC Portugal.
8. Chegini, V.; Aghtouman, P. Hydraulic stability formulae for Antifer blocks, 2006.
9. Freitas, P. M. G. *Hydraulic Stability of Antifer Cubes: Physical Model Study*. MSc dissertation, Instituto Superior Tecnico, 2013.

10. Scaravaglione, G.; Latham, J.-P.; Xiang, J.; Francone, A.; Tomasicchio, G. R. “Historical overview of the structural integrity of Concrete Armour Units”, 2022.
11. Edge, B. L.; Magoon, O. T.; et al. ASCE-ICCE documentation on the rehabilitation of the West Breakwater of Sines and the use of high-density Antifer cubes.
12. Reis, M. T.; Fortes, C. J. E. M.; Neves, M. G.; et al. “Rehabilitation of Sines west breakwater: wave overtopping study”, *Proceedings of the Institution of Civil Engineers - Maritime Engineering*, 2011.
13. Troch, P.; De Rouck, J.; Van Damme, L. “Full-scale wave-overtopping measurements on the Zeebrugge rubble mound breakwater”, *Coastal Engineering*, 2004.
14. Geeraerts, J.; De Rouck, J.; Troch, P.; et al. “Effects of new variables on the overtopping discharge at steep rubble mound breakwaters: the Zeebrugge prototype case”, 2009.
15. Jensen, O. J. Historical syntheses on the evolution of armour units and recent observations on Antifer behaviour on highly exposed breakwaters.
16. Luis, L.; Freire, S.; Barros, J.; Lopes, H.; Lemos, R.; Fortes, C.; Neves, G. “Estudos e projetos do prolongamento do quebra-mar exterior e das acessibilidades maritimas do Porto de Leixoes”, PIANC Portugal / LNEC, 2022.
17. Teixeira Duarte. “Extension of the Outer Breakwater and Maritime Access Ways of Leixoes Port”, project description with Antifer block quantities and classes.
18. Portos dos Acores / NRV-Norvia / PORTUS. Documentation on the emergency protection and reconstruction of the Port of Lajes das Flores after Hurricane Lorenzo and subsequent storm damage.